ECE 523 Quantum Computing Portfolio

Manny Serrano Campa

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Introduction

This portfolio contains reading notes, exercises and glossary terms related to Quantum Computing, developed throughout the Fall 2026 semester as part of my ECE 523 Quantum Computing course.

Exercises were taken from the Exercise Set pdf constructed by Prof. Natalie Klco. // insert link here.

Part I
Reading Notes

Part II
Exercises

I Warm-Up and Preliminaries

A Tensor Products

I.A.1 Parameters Needed to Specify a Quantum State

We count the independent real parameters needed to specify a pure physical state. Normalization constrains the amplitudes, and an overall phase does not distinguish physical states.

Starting with one qubit. A single qubit belongs to a two-dimensional Hilbert space and can be written as

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle\,, \qquad a, b \in \mathbb{C}.$$

Each complex amplitude contains two real numbers, so we begin with four real parameters. Normalization requires

$$|a|^2 + |b|^2 = 1,$$

which removes one independent real parameter because the total measurement probability must equal one.

An overall phase removes one more parameter. To see why, multiplying the entire state by $e^{i\gamma}$ leaves the probability of any projective measurement outcome unchanged:

$$\left|\langle u|\, e^{i\gamma}\, |\psi\rangle\right|^2 = \left|e^{i\gamma}\, \langle u|\psi\rangle\right|^2 = |\langle u|\psi\rangle|^2\,.$$

This freedom applies to the global phase; the relative phase between a and b can affect interference and must be retained. Therefore, one qubit requires

$$4 - 1 - 1 = 2$$

independent real parameters.

A separable state of n qubits. For a fully separable pure state,

$$|\Psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle \,,$$

we can specify each qubit individually. Since each qubit requires two real parameters, the total is

$$\boxed{N_{\text{separable}} = 2n.}$$

There are no additional physical parameters associated with the individual global phases: multiplying each factor by $e^{i\gamma_j}$ only multiplies the full product by the unobservable global phase $e^{i\sum_j \gamma_j}$.

A general state of n qubits. The combined Hilbert space has dimension 2^n. A general pure state can therefore be expanded as

$$|\Psi\rangle = \sum_{x=0}^{2^n-1} c_x \, |x\rangle \,,$$

where $|x\rangle$ labels a computational basis state using the n-bit binary representation of x.

There are 2^n complex amplitudes, giving $2 \cdot 2^n$ real parameters before imposing physical constraints. The normalization condition

$$\sum_{x=0}^{2^n-1} |c_x|^2 = 1$$

removes one parameter, and the global phase removes another. Thus,

$$\boxed{N_{\text{general}} = 2^{n+1} - 2.}$$

This is the parameter count for an unrestricted pure state, including a generic entangled state.

The difference in scaling comes from whether the joint amplitudes factor into individual qubit amplitudes. A separable state can be described one qubit at a time, so its parameter count grows linearly with n. A general state requires amplitudes for the full joint basis, so its parameter count grows exponentially.

d-level quantum degrees of freedom A single d-level system has d complex amplitudes. After removing normalization and global phase, it requires $2d - 2$ independent real parameters. For n fully separable d-level systems,

$$\boxed{N_{\text{separable}} = n(2d-2) = 2n(d-1).}$$

For a general joint pure state, the Hilbert space has dimension d^n. Applying the same counting argument gives

$$\boxed{N_{\text{general}} = 2d^n - 2.}$$

As a consistency check, setting $d = 2$ recovers the qubit results. Setting $n = 1$ makes both counts equal to $2d - 2$, as expected: with only one subsystem, there is no distinction between a product state and a general state arising from entanglement between subsystems.

I.A.2 Tensor Product of Three Identical Qubit States

The notation $|\psi\rangle^{\otimes 3}$ means taking the tensor product of $|\psi\rangle$ with itself three times. For

$$|\psi\rangle = \frac{|0\rangle + \sqrt{2}\,|1\rangle}{\sqrt{3}},$$

the joint state is

$$|\psi\rangle^{\otimes 3} = \frac{1}{3\sqrt{3}} \Big(|0\rangle + \sqrt{2}\,|1\rangle\Big) \otimes \Big(|0\rangle + \sqrt{2}\,|1\rangle\Big) \otimes \Big(|0\rangle + \sqrt{2}\,|1\rangle\Big).$$

The tensor product distributes over addition. Starting with the first two factors gives

$$|\psi\rangle^{\otimes 2} = \frac{1}{3} \Big(|00\rangle + \sqrt{2}\,|01\rangle + \sqrt{2}\,|10\rangle + 2\,|11\rangle\Big).$$

Taking the tensor product with the third qubit then gives

$$\begin{aligned} |\psi\rangle^{\otimes 3} = \frac{1}{3\sqrt{3}} \Big(&|000\rangle + \sqrt{2}\,|001\rangle + \sqrt{2}\,|010\rangle + 2\,|011\rangle \\ &+ \sqrt{2}\,|100\rangle + 2\,|101\rangle + 2\,|110\rangle + 2\sqrt{2}\,|111\rangle \Big). \end{aligned}$$

Each coefficient is the product of the three individual qubit amplitudes. Thus, each occurrence of a 1 contributes a factor of $\sqrt{2}$ inside the parentheses. For example, $|101\rangle$ has two ones, giving a coefficient of $(\sqrt{2})^2 = 2$.

Using the computational basis in binary order,

$$(|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle),$$

we obtain the column vector

$$\boxed{|\psi\rangle^{\otimes 3} = \frac{1}{3\sqrt{3}} \begin{pmatrix} 1 \\ \sqrt{2} \\ \sqrt{2} \\ 2 \\ \sqrt{2} \\ 2 \\ 2 \\ 2\sqrt{2} \end{pmatrix}}$$

There are eight entries because the three-qubit Hilbert space has dimension $2^3 = 8$. As a normalization check, the squared amplitudes sum to

$$\frac{1}{27}\,(1 + 2 + 2 + 4 + 2 + 4 + 4 + 8) = 1.$$

Although the expansion contains all eight computational basis states, the state is separable since it was constructed as a tensor product of three individual qubit states. A superposition of computational basis states does not, by itself, imply entanglement.

I.A.3 Tensor Products and Reordering Subsystems

The tensor product creates a larger vector space by combining the vector spaces of the individual systems. We can start with two vectors:

$$|a\rangle = \begin{pmatrix} a_0 \\ a_1 \end{pmatrix}, \qquad |b\rangle = \begin{pmatrix} b_0 \\ b_1 \end{pmatrix}.$$

To take their tensor product, we multiply each entry of $|a\rangle$ by the entire vector $|b\rangle$:

$$|a\rangle \otimes |b\rangle = \begin{pmatrix} a_0 |b\rangle \\ a_1 |b\rangle \end{pmatrix} = \begin{pmatrix} a_0 b_0 \\ a_0 b_1 \\ a_1 b_0 \\ a_1 b_1 \end{pmatrix}.$$

So two vectors with two entries each give a joint vector with four entries.

Implementing the tensor product. For vectors of lengths d_1 and d_2, we first create an output vector with $d_1 d_2$ entries. Then we go through each entry of the first vector and multiply it by each entry of the second vector, keeping that order.

Using indices starting at zero, this is

$$c_{id_2+j} = a_i b_j, \qquad 0 \le i < d_1, \quad 0 \le j < d_2.$$

The index i stays fixed while j goes through the second vector. Then we move to the next value of i.

For matrices, we do the same thing: replace each entry of A with that entry times the entire matrix B:

$$A \otimes B = \begin{pmatrix} A_{00}B & \cdots & A_{0,n-1}B \\ \vdots & \ddots & \vdots \\ A_{m-1,0}B & \cdots & A_{m-1,n-1}B \end{pmatrix}.$$

If A is $m \times n$ and B is $p \times q$, the result is an $mp \times nq$ matrix. Its entries are

$$(A \otimes B)_{ip+k,\, jq+\ell} = A_{ij} B_{k\ell}.$$

How the basis is ordered. For the two-qubit example, the entries correspond to

$$(|00\rangle, |01\rangle, |10\rangle, |11\rangle).$$

The rightmost bit changes first. This is the same as counting in binary from 0 to 3.

More generally, the basis state $|i\rangle_1 \otimes |j\rangle_2$ has index

$$x = id_2 + j.$$

We go through every state of the second space before moving to the next state of the first space. If both spaces use binary ordering, we join their bit strings to label the joint state.

Building the reordering matrices. To build each matrix, we look at where every input basis state goes. The input determines the column, and the output determines the row. We put a 1 at that position and zeros everywhere else.

If the input index x goes to $f(x)$, then

$$U = \sum_x |f(x)\rangle \langle x| .$$

Below, all row and column indices start at zero. A listed pair (r, c) means $U_{r,c} = 1$, so it also specifies the outer-product term $|r\rangle \langle c|$.

Reversing three qubits. For the order $(1, 2, 3) \to (3, 2, 1)$, we swap the first and third bits:

$$U_{\text{rev}} |abc\rangle = |cba\rangle .$$

The input index is $4a + 2b + c$, and the output index is $4c + 2b + a$. For example,

$$|001\rangle \to |100\rangle ,$$

so column 1 has its 1 in row 4.

Applying this to all eight basis states gives

$$U_{\text{rev}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} .$$

The eight nonzero entries are

$$(r, c) \in \{(0, 0), (4, 1), (2, 2), (6, 3), \\ (1, 4), (5, 5), (3, 6), (7, 7)\}.$$

Reordering four qubits. For the order $(1, 2, 3, 4) \to (2, 3, 1, 4)$, the basis states change as

$$U_{\text{cycle}} |abcd\rangle = |bcad\rangle .$$

The input and output indices are

$$x_{\text{in}} = 8a + 4b + 2c + d, \qquad x_{\text{out}} = 8b + 4c + 2a + d.$$

For example,

$$|1010\rangle \to |0110\rangle ,$$

which gives a 1 in row 6, column 10.

Since the fourth bit stays the same, we can write the 16×16 matrix using 2×2 blocks:

$$U_{\text{cycle}} = \begin{pmatrix} I & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & I & 0 & 0 & 0 \\ 0 & I & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I & 0 & 0 \\ 0 & 0 & I & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & I & 0 \\ 0 & 0 & 0 & I & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & I \end{pmatrix}, \qquad I = \mathbb{I}_2, \quad 0 = 0_{2\times 2}.$$

Each identity block keeps the fourth bit unchanged. The sixteen nonzero entries are

$$\begin{aligned}(r, c) \in \{&(0,0), (1,1), (4,2), (5,3), \\ &(8,4), (9,5), (12,6), (13,7), \\ &(2,8), (3,9), (6,10), (7,11), \\ &(10,12), (11,13), (14,14), (15,15)\}.\end{aligned}$$

Swapping two three-level systems. Now each subsystem has three basis states, labeled 0, 1, and 2. The joint basis is

$$(|00\rangle, |01\rangle, |02\rangle, |10\rangle, |11\rangle, |12\rangle, |20\rangle, |21\rangle, |22\rangle).$$

We want

$$U_{\text{swap}} |ab\rangle = |ba\rangle .$$

The input index is $3a + b$, and the output index is $3b + a$. This gives

$$U_{\text{swap}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

The nine nonzero entries are

$$\begin{aligned}(r, c) \in \{&(0,0), (3,1), (6,2), \\ &(1,3), (4,4), (7,5), \\ &(2,6), (5,7), (8,8)\}.\end{aligned}$$

Checking the results. Each matrix has exactly one 1 in every row and column. This means its columns are normalized and orthogonal, so

$$U^\dagger U = \mathbb{I}.$$

Thus, all three transformations are unitary. They rearrange the amplitudes while preserving the total probability.

We can also check how many times we must apply each transformation to get back to the original state. Reversing three qubits twice or swapping two qutrits twice restores the original order:

$$U_{\text{rev}}^2 = \mathbb{I}_8, \qquad U_{\text{swap}}^2 = \mathbb{I}_9.$$

The four-qubit transformation cycles the first three positions, so it takes three applications:

$$U_{\text{cycle}}^3 = \mathbb{I}_{16}.$$

C Hermitian and Unitary Matrices

I.C.2 Time Evolution for a Time-Independent Hamiltonian

We start with the Schrödinger equation,

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle ,$$

where $\hat{H}$ does not depend on time. We want to find the operator that takes an initial state at time t_0 to its state at time t:

$$|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle .$$

Substituting this into the Schrödinger equation gives

$$i\hbar \frac{\partial U(t, t_0)}{\partial t} |\psi(t_0)\rangle = \hat{H} U(t, t_0) |\psi(t_0)\rangle .$$

The initial state $|\psi(t_0)\rangle$ is fixed, so the time derivative acts only on $U(t, t_0)$. Since this equation must hold for any initial state, the operators must satisfy

$$\frac{\partial U(t, t_0)}{\partial t} = -\frac{i}{\hbar} \hat{H} U(t, t_0).$$

This has the same form as the differential equation $\frac{df}{dt} = af$ with constant a, whose solution is an exponential. Here, the constant is the operator $-i\hat{H}/\hbar$. We also need

$$U(t_0, t_0) = \mathbb{I},$$

because evolving through zero time must leave the state unchanged. Together, these give

$$\boxed{U(t, t_0) = e^{-\frac{i}{\hbar}\hat{H}(t-t_0)}.}$$

Therefore, the state at time t is

$$\boxed{|\psi(t)\rangle = e^{-\frac{i}{\hbar}\hat{H}(t-t_0)} |\psi(t_0)\rangle .}$$

Checking the solution. Since $\hat{H}$ is time independent, differentiating the matrix exponential gives

$$\frac{\partial U}{\partial t} = -\frac{i}{\hbar} \hat{H} e^{-\frac{i}{\hbar}\hat{H}(t-t_0)} = -\frac{i}{\hbar} \hat{H} U,$$

which is the equation we needed to solve. At $t = t_0$, the exponent is zero, so $U(t_0, t_0) = \mathbb{I}$.

For a short time interval Δt, the expansion is

$$U(t_0 + \Delta t, t_0) = \mathbb{I} - \frac{i}{\hbar} \hat{H} \Delta t + \mathcal{O}(\Delta t^2).$$

This shows directly how the Hamiltonian determines the change in the state over a small amount of time.

What this means physically. If the initial state is an energy eigenstate, $\hat{H}\,|E_n\rangle = E_n\,|E_n\rangle$, then

$$U(t,t_0)\,|E_n\rangle = e^{-\frac{i}{\hbar}E_n(t-t_0)}\,|E_n\rangle\,.$$

It only gains a global phase, so its measurement probabilities do not change.

For a superposition of energy eigenstates,

$$|\psi(t_0)\rangle = \sum_n c_n\,|E_n\rangle\,,$$

each component gains its own phase:

$$|\psi(t)\rangle = \sum_n c_n e^{-\frac{i}{\hbar}E_n(t-t_0)}\,|E_n\rangle\,.$$

The energy probabilities $|c_n|^2$ stay the same, but the relative phases between components with different energies change. These relative phases can change interference and the probabilities of measurements in other bases.

Finally, because $\hat{H}$ is Hermitian, the time-evolution operator is unitary:

$$U^\dagger(t,t_0)U(t,t_0) = \mathbb{I}.$$

Thus, the evolution preserves normalization and the total measurement probability remains one.

I.C.3 The Exponential of a Hermitian Matrix

We are given that A is Hermitian, so $A^\dagger = A$. To show that e^{iA} is unitary, we need to show

$$(e^{iA})^\dagger e^{iA} = \mathbb{I}.$$

First, we write the exponential as a power series:

$$e^{iA} = \mathbb{I} + iA + \frac{(iA)^2}{2!} + \frac{(iA)^3}{3!} + \cdots\,.$$

Taking the Hermitian conjugate conjugates the scalar coefficients and transposes and conjugates the matrices. Since $i^* = -i$ and $A^\dagger = A$, we have

$$(iA)^\dagger = -iA.$$

For powers of the same matrix,

$$\big((iA)^n\big)^\dagger = \big((iA)^\dagger\big)^n = (-iA)^n.$$

Although taking the Hermitian conjugate reverses the order of a product, all the factors here are identical.

Applying this to the series gives

$$\begin{aligned}(e^{iA})^\dagger &= \mathbb{I} - iA + \frac{(-iA)^2}{2!} + \frac{(-iA)^3}{3!} + \cdots \\ &= e^{-iA}.\end{aligned}$$

Now we substitute this into the unitarity condition:

$$(e^{iA})^\dagger e^{iA} = e^{-iA}e^{iA}.$$

The matrices $-iA$ and iA commute because they are scalar multiples of the same matrix. Therefore, we can combine their exponentials:

$$e^{-iA}e^{iA} = e^{-iA+iA} = e^0 = \mathbb{I}.$$

The reverse product gives the identity as well, so

$$\boxed{(e^{iA})^\dagger e^{iA} = e^{iA}(e^{iA})^\dagger = \mathbb{I}.}$$

Thus, e^{iA} is unitary for every Hermitian matrix A.

Physically, this connects Hermitian Hamiltonians with unitary time evolution. For a time-independent Hamiltonian H, the operator

$$U(t) = e^{-iHt/\hbar}$$

has this form with $A = -Ht/\hbar$, which is Hermitian for real t. Its unitarity preserves the norm:

$$\langle\psi(t)|\psi(t)\rangle = \langle\psi(0)|\, U^\dagger(t)U(t)\,|\psi(0)\rangle = \langle\psi(0)|\psi(0)\rangle\,.$$

So a normalized initial state stays normalized, as required for the total measurement probability to remain one.

IV Toward Quantum Simulation

B Projecting Spatial Parity

Swapping the two qubits. Interchanging the two Hilbert spaces means swapping the states occupying the first and second qubit positions. We represent this with the SWAP operator:

$$\hat{S}\,|ab\rangle = |ba\rangle\,.$$

On the computational basis states, this gives

$$\hat{S}\,|00\rangle = |00\rangle\,, \qquad \hat{S}\,|01\rangle = |10\rangle\,, \qquad \hat{S}\,|10\rangle = |01\rangle\,, \qquad \hat{S}\,|11\rangle = |11\rangle\,.$$

Since the operator is linear, we can apply it separately to each term in a Bell state.

The three triplet states. For the first state,

$$\begin{aligned}\hat{S}\,|\psi_{t,1}\rangle &= \hat{S}\frac{|00\rangle + |11\rangle}{\sqrt{2}} \\ &= \frac{|00\rangle + |11\rangle}{\sqrt{2}} = |\psi_{t,1}\rangle\,.\end{aligned}$$

Both terms stay the same under the swap.

For the second state,

$$\begin{aligned}\hat{S}\,|\psi_{t,2}\rangle &= \hat{S}\frac{|01\rangle + |10\rangle}{\sqrt{2}} \\ &= \frac{|10\rangle + |01\rangle}{\sqrt{2}} = |\psi_{t,2}\rangle\,.\end{aligned}$$

The terms exchange places, but their sum is unchanged.

For the third state,

$$\begin{aligned}\hat{S}\left|\psi_{t,3}\right\rangle &= \hat{S}\frac{|00\rangle - |11\rangle}{\sqrt{2}} \\ &= \frac{|00\rangle - |11\rangle}{\sqrt{2}} = \left|\psi_{t,3}\right\rangle .\end{aligned}$$

Again, each basis state stays in place. The minus sign between the terms does not make this state antisymmetric, because swapping the qubits does not exchange these two terms.

Therefore, all three triplet states satisfy

$$\boxed{\hat{S}\left|\psi_{t,j}\right\rangle = +\left|\psi_{t,j}\right\rangle , \qquad j = 1, 2, 3.}$$

They are symmetric under exchange and have positive spatial parity.

The singlet state. For the remaining Bell state,

$$\begin{aligned}\hat{S}\left|\psi_{s}\right\rangle &= \hat{S}\frac{|01\rangle - |10\rangle}{\sqrt{2}} \\ &= \frac{|10\rangle - |01\rangle}{\sqrt{2}} \\ &= -\frac{|01\rangle - |10\rangle}{\sqrt{2}} = -\left|\psi_{s}\right\rangle .\end{aligned}$$

Here, exchanging the two terms introduces an overall minus sign. Thus,

$$\boxed{\hat{S}\left|\psi_{s}\right\rangle = -\left|\psi_{s}\right\rangle .}$$

The singlet is antisymmetric under exchange and has negative spatial parity.

These results give three independent states with SWAP eigenvalue $+1$ and one with eigenvalue -1, accounting for all four dimensions of the two-qubit Hilbert space. Here, parity refers to symmetry under exchanging the qubits.

C Time Evolution and Error Mitigation

IV.C.1 Quantum Zeno Effect

Our initial state is $|\psi(0)\rangle = |0\rangle$, and the Hamiltonian is

$$\hat{H} = \frac{X + Y}{\sqrt{2}}.$$

Using the units in the question, the time-evolution operator is

$$U_{\text{ext}}(t) = e^{-i\hat{H}t}.$$

We want to find how frequently we need to measure in the Z basis to keep the state near $|0\rangle$ over one unit of time.

Evolution between measurements. First, we calculate the square of the Hamiltonian:

$$\hat{H}^2 = \frac{(X+Y)(X+Y)}{2} = \frac{X^2 + XY + YX + Y^2}{2}.$$

Since $X^2 = Y^2 = \mathbb{I}$ and $XY + YX = 0$, we get

$$\hat{H}^2 = \mathbb{I}.$$

Therefore, the even powers in the exponential series give the cosine terms, while the odd powers give the sine terms:

$$e^{-i\hat{H}t} = \mathbb{I}\cos t - i\hat{H}\sin t.$$

After a time interval Δt, the state becomes

$$\begin{aligned} |\psi(\Delta t)\rangle &= U_{\text{ext}}(\Delta t)\,|0\rangle \\ &= \cos(\Delta t)\,|0\rangle - i\frac{X+Y}{\sqrt{2}}\sin(\Delta t)\,|0\rangle\,. \end{aligned}$$

Using $X\,|0\rangle = |1\rangle$ and $Y\,|0\rangle = i\,|1\rangle$,

$$\begin{aligned} |\psi(\Delta t)\rangle &= \cos(\Delta t)\,|0\rangle - i\frac{1+i}{\sqrt{2}}\sin(\Delta t)\,|1\rangle \\ &= \cos(\Delta t)\,|0\rangle + \frac{1-i}{\sqrt{2}}\sin(\Delta t)\,|1\rangle\,. \end{aligned}$$

Probability of measuring zero. Measuring in the Z basis gives either $|0\rangle$ or $|1\rangle$. The probability of getting $|0\rangle$ is

$$p_0(\Delta t) = |\langle 0|\psi(\Delta t)\rangle|^2 = \cos^2(\Delta t).$$

If the result is zero, the measurement projects the state back to $|0\rangle$. The next interval then starts from the same state as the previous one.

For N equally spaced measurements over a total time $T = 1$, the interval is

$$\Delta t = \frac{T}{N} = \frac{1}{N}.$$

Here, the measurements occur at $\Delta t, 2\Delta t, \ldots, N\Delta t = 1$; the initial preparation is not counted as a measurement.

To get zero at every measurement, we multiply the conditional probability for each interval:

$$\boxed{P_{\text{survival}}(N) = \left[\cos^2\left(\frac{1}{N}\right)\right]^N.}$$

This counts runs in which every result is zero, rather than runs that might leave $|0\rangle$ and return to it before the final measurement.

Finding the number of measurements. We need

$$\left[\cos^2\left(\frac{1}{N}\right)\right]^N \geq 0.99.$$

For large N, we can use the small-angle approximation $\cos^2 x \approx 1 - x^2$:

$$P_{\text{survival}}(N) \approx \left(1 - \frac{1}{N^2}\right)^N \approx 1 - \frac{1}{N}.$$

The second approximation keeps the leading correction in the binomial expansion. This gives

$$1 - \frac{1}{N} \gtrsim 0.99,$$

so we expect to need about 100 measurements.

Checking the exact expression at the two nearby integers gives

$$P_{\text{survival}}(99) \approx 0.98994966 < 0.99,$$

whereas

$$P_{\text{survival}}(100) \approx 0.99004967 > 0.99.$$

The survival probability increases with N for this equally spaced protocol. Thus, the minimum is

$$\boxed{N_{\min} = 100, \qquad \Delta t = 0.01.}$$

Why repeated measurements help. For a short interval, the probability of leaving $|0\rangle$ is

$$1 - p_0(\Delta t) = \sin^2(\Delta t) \approx (\Delta t)^2.$$

Making the intervals shorter reduces this probability quadratically. Even though we perform more measurements, the total failure probability scales approximately as

$$N(\Delta t)^2 = N\left(\frac{1}{N}\right)^2 = \frac{1}{N}.$$

It therefore approaches zero as N increases. This is the quantum Zeno effect: frequent measurements suppress transitions away from the initial state.

The state still evolves slightly between measurements. For $N = 100$, along a run with all zero outcomes, its overlap probability with $|0\rangle$ stays at least $\cos^2(0.01) \approx 0.999900$ during each interval. The measurements repeatedly return it to $|0\rangle$, keeping it in that state's vicinity.

IV.C.2 Symmetry Identification

Our initial state is

$$|\psi(0)\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = |\Phi^+\rangle\,,$$

and the time-evolution operator is

$$U_{\text{noiseless}}(t) = e^{-i(X\otimes Y)t}.$$

We use the Bell states

$$|\Phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \qquad |\Psi^\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}.$$

(a) Evolution in the Bell basis. Since

$$(X \otimes Y)^2 = X^2 \otimes Y^2 = \mathbb{I}_4,$$

we can rewrite the evolution operator as

$$U_{\text{noiseless}}(t) = \mathbb{I}_4 \cos t - i(X \otimes Y) \sin t.$$

First, we apply the Hamiltonian to the computational basis states in $|\Phi^+\rangle$:

$$(X \otimes Y)\,|00\rangle = i\,|11\rangle\,, \qquad (X \otimes Y)\,|11\rangle = -i\,|00\rangle\,.$$

Therefore,

$$\begin{aligned}(X \otimes Y)\,|\Phi^+\rangle &= \frac{i\,|11\rangle - i\,|00\rangle}{\sqrt{2}} \\ &= -i\frac{|00\rangle - |11\rangle}{\sqrt{2}} \\ &= -i\,|\Phi^-\rangle\,.\end{aligned}$$

The state after time t is then

$$\begin{aligned}|\psi(t)\rangle &= U_{\text{noiseless}}(t)\,|\Phi^+\rangle \\ &= \cos t\,|\Phi^+\rangle - i(-i)\sin t\,|\Phi^-\rangle \\ &= \cos t\,|\Phi^+\rangle - \sin t\,|\Phi^-\rangle\,.\end{aligned}$$

Since the Bell basis is orthonormal, the measurement probabilities are

$$\boxed{\begin{aligned}&P_{\Phi^+}(t) = \cos^2 t, \\ &P_{\Phi^-}(t) = \sin^2 t, \\ &P_{\Psi^+}(t) = 0, \\ &P_{\Psi^-}(t) = 0.\end{aligned}}$$

The probabilities sum to

$$\cos^2 t + \sin^2 t = 1.$$

The state evolves between the two Φ Bell states, while the two Ψ Bell states remain unpopulated.

(b) Identifying the symmetry operator. We need to find what the populated states have in common. Both $|\Phi^+\rangle$ and $|\Phi^-\rangle$ contain only $|00\rangle$ and $|11\rangle$, so the operator $Z \otimes Z$ works.

Using $Z\,|0\rangle = |0\rangle$ and $Z\,|1\rangle = -\,|1\rangle$,

$$\begin{aligned}(Z \otimes Z)\,|\Phi^+\rangle &= \frac{|00\rangle + |11\rangle}{\sqrt{2}} = +\,|\Phi^+\rangle\,, \\ (Z \otimes Z)\,|\Phi^-\rangle &= \frac{|00\rangle - |11\rangle}{\sqrt{2}} = +\,|\Phi^-\rangle\,.\end{aligned}$$

For $|11\rangle$, the two minus signs cancel. Thus, both populated states have eigenvalue $+1$.

For the other two Bell states,

$$\begin{aligned}(Z \otimes Z)\,|\Psi^+\rangle &= \frac{-\,|01\rangle - |10\rangle}{\sqrt{2}} = -\,|\Psi^+\rangle\,, \\ (Z \otimes Z)\,|\Psi^-\rangle &= \frac{-\,|01\rangle + |10\rangle}{\sqrt{2}} = -\,|\Psi^-\rangle\,.\end{aligned}$$

These states have eigenvalue -1. Therefore, $Z \otimes Z$ distinguishes the populated states from those that remain unpopulated:

$$\boxed{\langle Z \otimes Z \rangle_{\Phi^\pm} = +1, \qquad \langle Z \otimes Z \rangle_{\Psi^\pm} = -1.}$$

Here, parity refers to whether the two computational basis bits are equal or different.

(c) Showing that parity is conserved. With $\hat{H} = X \otimes Y$, the commutator is

$$\begin{aligned}[Z \otimes Z, \hat{H}] &= (Z \otimes Z)(X \otimes Y) - (X \otimes Y)(Z \otimes Z) \\ &= ZX \otimes ZY - XZ \otimes YZ.\end{aligned}$$

Since $ZX = -XZ$ and $ZY = -YZ$,

$$\begin{aligned}[Z \otimes Z, \hat{H}] &= (-XZ) \otimes (-YZ) - XZ \otimes YZ \\ &= (-1)(-1)(XZ \otimes YZ) - XZ \otimes YZ \\ &= 0.\end{aligned}$$

Therefore,

$$\boxed{[Z \otimes Z, \hat{H}] = 0.}$$

For a time-independent observable, with $\hbar = 1$,

$$\frac{d}{dt}\langle A \rangle = i\langle [\hat{H}, A] \rangle.$$

Applying this to $Z \otimes Z$ gives

$$\frac{d}{dt}\langle Z \otimes Z \rangle = -i\langle [Z \otimes Z, \hat{H}] \rangle = 0.$$

The initial state has parity $+1$, so

$$\boxed{\langle Z \otimes Z \rangle_t = +1 \quad \text{for all } t.}$$

The state can change between $|\Phi^+\rangle$ and $|\Phi^-\rangle$, but its parity stays the same. This is the conserved symmetry under the time evolution.

IV.C.3 Coherent Errors: Zeno and Basis Randomization

Now there is presence of single-qubit errors such that the time evolution becomes:

$$U_{\text{noisy}}(t) = e^{-iH_{\text{noisy}}t}, \qquad H_{\text{noisy}} = X \otimes Y + I \otimes \frac{X+Y}{\sqrt{2}} + \frac{X+Y}{\sqrt{2}} \otimes I.$$

(a) Plotting Evolution of Probabilities We use the ordered Bell basis

$$\left\{ |\Phi^+\rangle, |\Phi^-\rangle, |\Psi^+\rangle, |\Psi^-\rangle \right\}.$$

In this basis, the noisy Hamiltonian is

$$H_{\text{noisy}} = \begin{pmatrix} 0 & i & \sqrt{2} & 0 \\ -i & 0 & -i\sqrt{2} & 0 \\ \sqrt{2} & i\sqrt{2} & 0 & -i \\ 0 & 0 & i & 0 \end{pmatrix}.$$

Starting in $|\Phi^+\rangle$, the initial state vector is

$$\psi(0) = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

For each sampled time t, we numerically calculate the full matrix exponential using Python's `scipy.linalg.expm`, then evolve the state:

$$\psi(t) = e^{-iH_{\text{noisy}}t}\psi(0).$$

The entries of $\psi(t)$ are the Bell-state amplitudes, so their squared magnitudes give the measurement probabilities:

$$P_j^{\text{noisy}}(t) = |\psi_j(t)|^2.$$

For comparison, the noiseless evolution found prior gives

$$P_{\Phi^+}^{\text{ideal}}(t) = \cos^2 t, \qquad P_{\Phi^-}^{\text{ideal}}(t) = \sin^2 t, \qquad P_{\Psi^\pm}^{\text{ideal}}(t) = 0.$$

Interpreting the $\pm 1\%$ tolerance as an absolute probability difference of 0.01, we calculate

$$\epsilon(t) = \max_j \left| P_j^{\text{noisy}}(t) - P_j^{\text{ideal}}(t) \right|.$$

All four probabilities remain within tolerance while $\epsilon(t) \leq 0.01$.

Using a time grid with spacing $\Delta t = 0.0001$, the first crossing of this tolerance occurs between $t = 0.0710$ and $t = 0.0711$. Thus, the probabilities remain within tolerance from $t = 0$ until approximately

$$\boxed{t_{\text{limit}} \approx 0.0710.}$$

Part III
Glossary

Glossary Entries

Acknowledgments